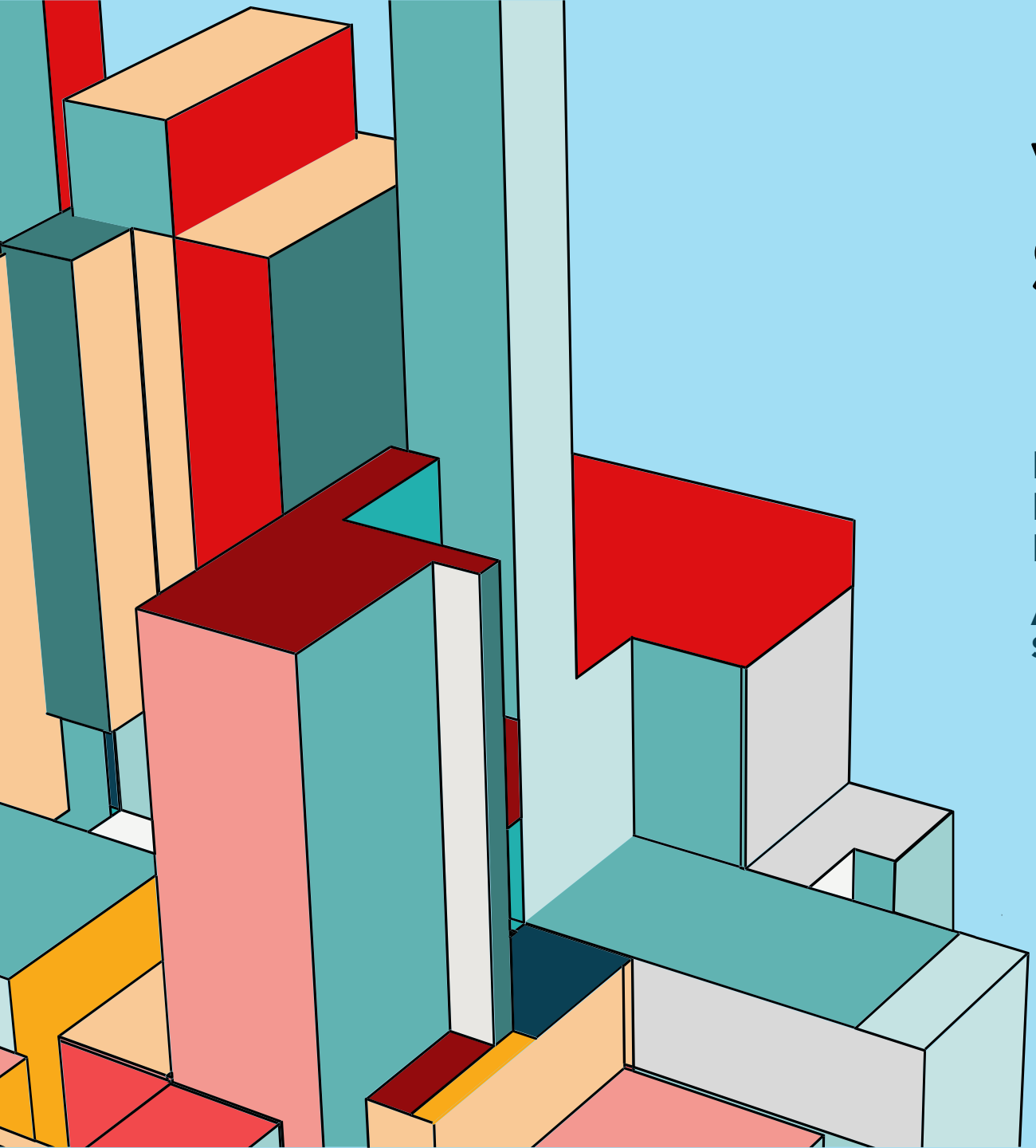


WILDFIRE SMOKE SEGMENTATION FROM SPACE

FROM GOES (NORTH AMERICA) TO METEOSAT-12 (EUROPE): CAN A SMOKE MODEL CROSS THE ATLANTIC?

ADSP 2026/27 · LINKS FOUNDATION – AI, DATA & SPACE

Politecnico di Torino

FONDAZIONE links PASSION FOR INNOVATION

ellis European Laboratory for Learning and Intelligent Systems

VALUE-DRIVEN PROJECT

Why this project?

- **Smoke travels far beyond the fire**: it drives air-quality alerts and health impacts hundreds of km downwind.
- **Europe has no expert smoke maps**: in the US, NOAA analysts draw smoke polygons daily (160k+ now public as SmokeViz). No European equivalent exists.
- **New opportunity**: Europe's new geostationary imager (MTG-FCI, Meteosat-12) sees the continent every 10 min with GOES-like bands.
- **Hard, open CV problem**: smoke is semi-transparent, diffuse and easily confused with clouds.

Stakeholders

Civil protection & emergency responders

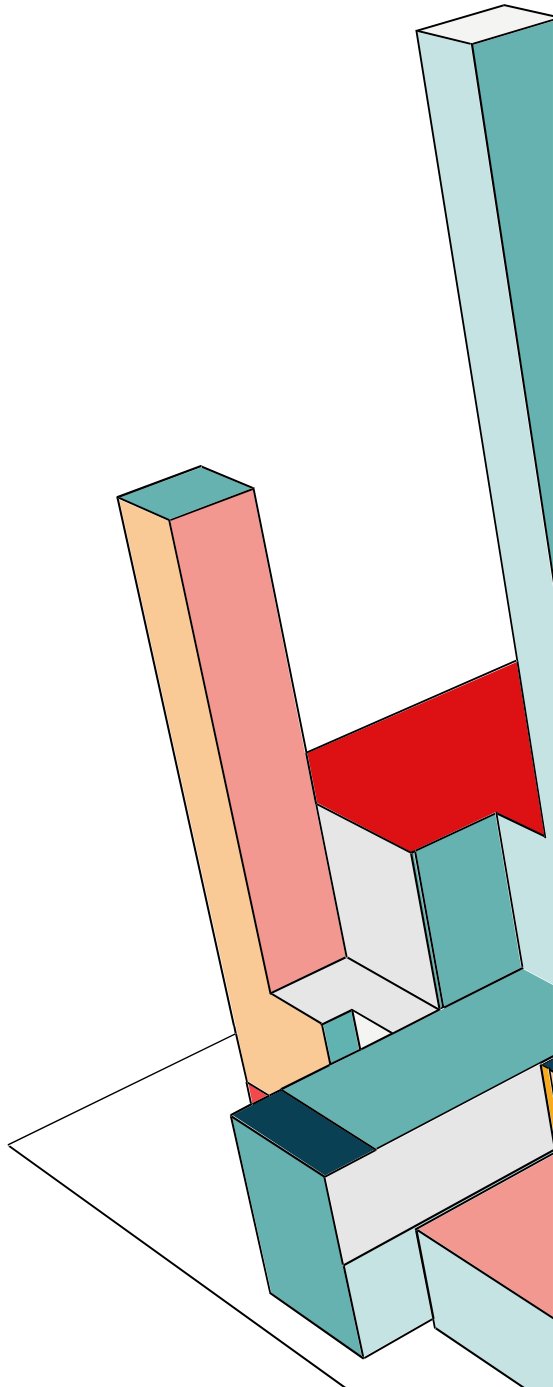
Air-quality / public-health agencies (e.g. regional ARPAs)

EUMETSAT & Copernicus user community

LINKS wildfire R&D (EU projects, follow-up M.Sc. thesis)

UN SDGs

- SDG 3 Good health — 3.9 air pollution
- SDG 11 Sustainable cities — 11.5 disaster losses
- SDG 13 Climate action — 13.1 resilience to hazards
- SDG 15 Life on land — 15.1 forest ecosystems



DATA

SmokeViz
main training set · NeurIPS 2025

160k+ smoke annotations by NOAA analysts on GOES ABI imagery, 2018–2023

256×256 GeoTIFF tiles + Light / Medium / Heavy density masks

Open on NOAA AWS S3; 1.2k-tile sample on Zenodo (1.3 GB, CC-BY 4.0)

GOES-16 Smoke Plumes
warm-up set · Kaggle

Small, ready-to-use image / mask set (GOES-16, Mexico)

First baseline in week 1–2; debug the pipeline before scaling up

Meteosat-12 MTG-FCI
transfer test · Europe

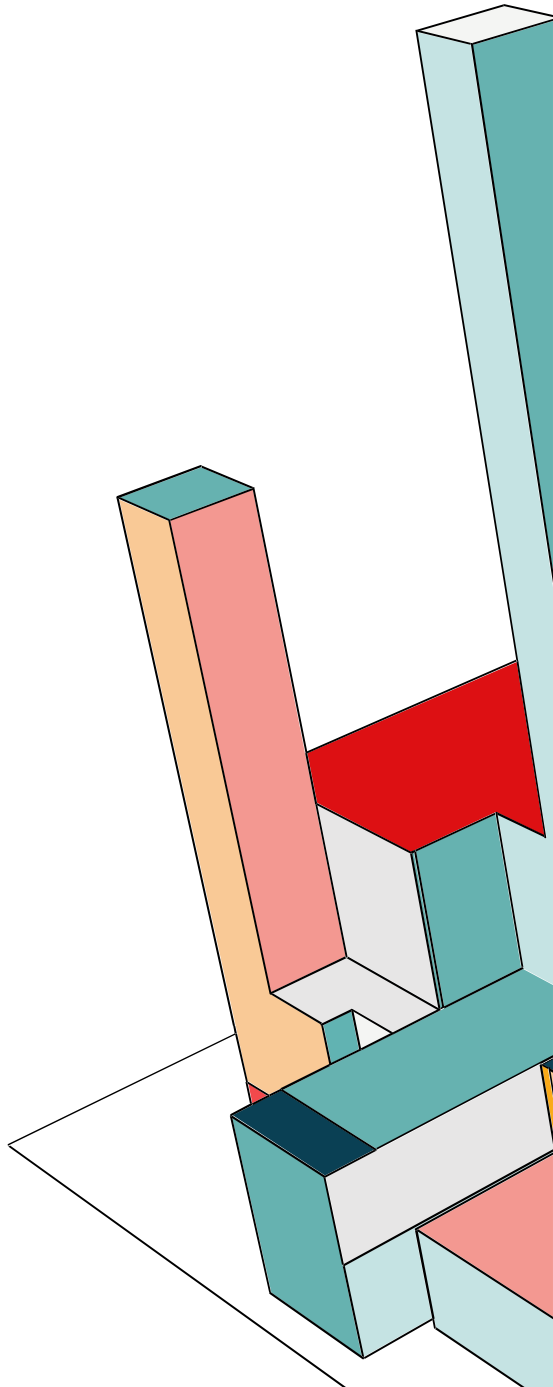
16 channels, 0.5–2 km, full disk every 10 min

Free L1c data from EUMETSAT Data Store, read with Satpy

Fire events picked from EFFIS / Copernicus EMS

How is it created and maintained?
NOAA Hazard Mapping System analysts hand-draw smoke polygons on GOES loops every day. SmokeViz rasterises them and uses pseudo-label dimension reduction (PLDR) to pick the GOES frame that best matches each annotation.

Is there a quality process?
Expert labels, but coarse polygons and subjective density classes, so noisy. Students audit label noise on a sample, hold out a full year for testing and hand-label ~30–50 European scenes as the transfer test set.



TASK

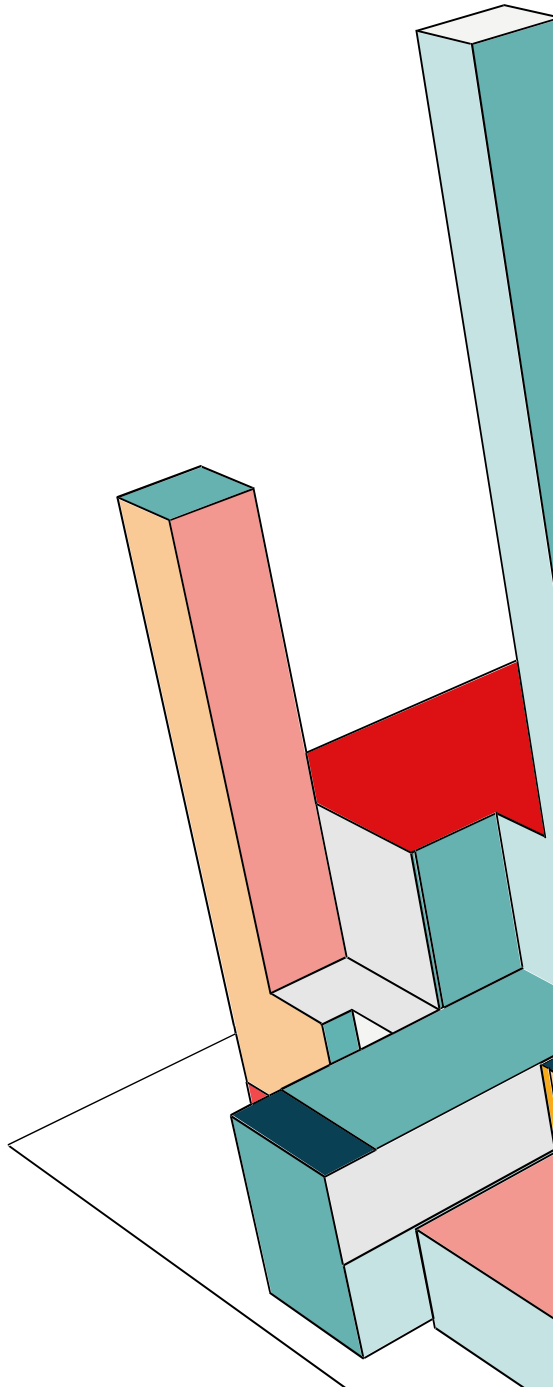
Semantic segmentation: per-pixel smoke density (none / light / medium / heavy) from multispectral geostationary tiles. Metrics: IoU and F1 per density class, plus smoke-vs-cloud false positives.

Research question: does a model trained on US GOES data work on European MTG imagery out of the box?

Wk 1–2 Data & EDA	Wk 3–5 GOES baselines	Wk 6–8 Transfer to MTG	Wk 9–10 Analysis & report
S3 subset (1–2 years), band stats, label-noise audit, Kaggle warm-up baseline	U-Net / SegFormer / etc., pretrained encoders, class-imbalance losses	ABI→FCI band mapping, 3–5 EU fires, zero-shot inference, small labelled test set	Error analysis (clouds, thin smoke, view angle), repo, final report

Deliverables: public GitHub repo (reproducible train/eval), baseline weights, small labelled MTG-FCI test set, short report.

Stretch goals: simple adaptation (per-band normalisation, few-shot fine-tuning), using FCI's extra green band, temporal stacking.



LIGHT MENTORING

- **Mentors**: Edoardo Arnaudo and Andrea Bragagnolo, LINKS Foundation – AI, Data & Space
- **Cadence**: 30-min call every two weeks, weekly around the kick-off and the transfer phase (wk 6)
- **Kick-off**: pointers to data access (NOAA S3, EUMETSAT Data Store), reading tools (rasterio, Satpy) and a short remote-sensing primer
- **Compute**: 256×256 tiles on a data subset, so a single GPU is enough
- **Open by design**: all data openly licensed, so the repo can be public from day one

Ideal team

3–4 students

PyTorch, computer vision / segmentation basics

No remote-sensing background needed

What's next

Strong results feed LINKS M.Sc. theses **"Multi-Sensor Wildfire Smoke Delineation"** (links-ads.github.io/proposals)

